GAT-CTDE: A Correctness-Aware Graph-Attention Architecture for Federated Multi-Agent Slice Admission Control

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Abstract

This paper delivers a working realisation of a graph-attention, multi-agent, federated slice admission controller, GAT-CTDE, treating correctness-aware evaluation as a the primary contribution alongside the architecture. This realisation closed three research gaps identified from our prior systematic review. On the metric that reflects the actual training objective, GAT-CTDE decisively beats an independent-learner ablation, though its edge over a single-agent baseline is small and not statistically decisive at this sample size. Scaled from 3 to 19 gNodeBs (gNBs) under a focused stress regime, GAT-CTDE resists a training collapse, demonstrating its advantage over the single-agent baseline, which completely collapsed. The GAT-CTDE model converged across 64 independently-drawn seeds on a residual 35.4% [25.5%, 45.8%] collapse rate already present at 7 gNBs. Additionally, privacy-preserving federated learning costs nothing measurable in reward and differential privacy degrades decision precision only past a real threshold, while disruption cost accelerates with severity for every arm. Two safeguards make these results trustworthy, namely a correctness-aware metric pair (44 of 49 independent checks disagreeing with the standard compliance metric) and a reproduction-versus-replication discipline. A live single-gNB deployment confirms the offline-trained decision logic executes correctly on real OpenAirInterface (OAI) hardware. During cross-layer integrity checks, a scheduler-configuration demonstrated that this rig cannot reach genuine per-slice scarcity at any tested load, so the live result establishes deployability rather than admission-control behaviour under contention, tested instead in the disclosed offline stress regime.

Keywords: Network Slicing, Slice Admission Control, Multi-Agent Reinforcement Learning, Graph Attention Networks, Federated Learning, Open RAN

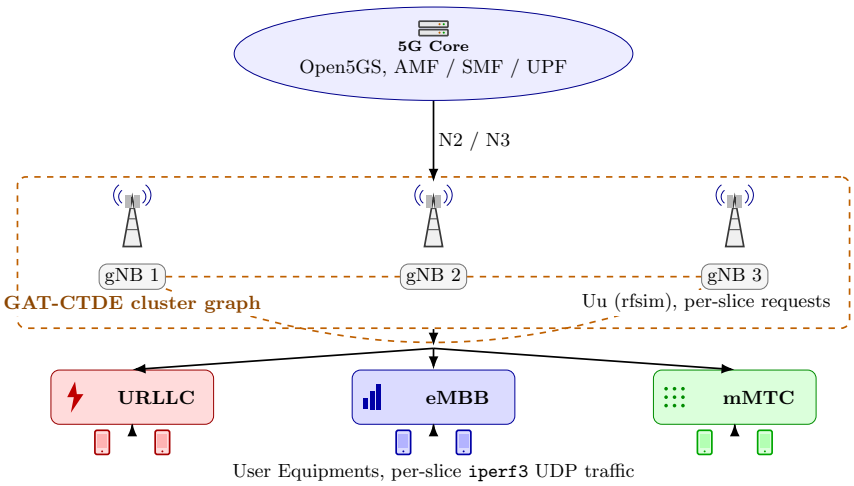


Fig. 1 System view. A cluster of gNBs, coordinated by the shared GAT-CTDE encoder over the cluster graph and backhauled to a 5G core, serves UEs across the URLLC, eMBB, and mMTC slices. Each gNB admits or rejects per-slice requests through a PRB-ratio ceiling.

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1 Introduction

Network slicing lets a single physical radio access network serve heterogeneous service classes, namely Ultra-Reliable Low-Latency Communication (URLLC), enhanced Mobile Broadband (eMBB), and massive Machine-Type Communication (mMTC), under distinct per-slice Service Level Agreements (SLAs). Our earlier work [7, 8] established that a Deep Q-Network (DQN) admission controller can learn to enforce these SLAs under a single gNodeB (gNB), validated on a live OpenAirInterface (OAI) testbed. A real multi-cell deployment does not admit requests at one gNB in isolation, since neighbouring cells share backhaul, spectrum planning, and a cluster-wide load-balance objective, making coordinated admission a genuinely multi-agent problem. This necessitates the investigation of a shared, graph-structured representation of the cluster's state to determine whether a set of per-gNB agents make better joint decisions than a single agent using a flattened joint state, or independent per-gNB agents with no shared representation. This research question motivated the premise for this paper. Fig. 1 places this setting in context.

This paper's contribution is GAT-CTDE, a centralised-training, decentralised-execution multi-agent architecture built on a shared Graph Attention Network (GAT) [17] encoder over the gNB topology feeding decentralised per-slice Q-heads, delivered together with the evaluation discipline needed to trust what it reports. Three gaps motivate this pairing, each named directly against our own systematic review of 67 core studies (Section 2). First, no reviewed study checks whether its

own outcome metric can score a degenerate policy no worse than a correct one. Second, none replicates a headline architectural finding on an independent seed sample before reporting it. Third, none tests whether a live testbed can actually reach the contention regime its offline claims are evaluated under. We close all three gaps, beginning with a per-component realisation of the architecture (GAT encoder, Multi-Agent Reinforcement Learning (MARL) controller, Federated Learning (FL) coordinator, Differential Privacy (DP) layer), each with its own measured, isolated contribution rather than one end-to-end number, a correctness-aware metric pair and reproduction-versus-replication discipline applied throughout rather than asserted once, and a live single-gNB deployment that tests, rather than assumes, whether this rig's own achievable load ever produces genuine per-slice scarcity.

Extended to federated training with differential privacy and evaluated under mid-episode gNB dropout, demand spikes, and agent churn, we scale the architecture to cluster sizes $N \in \{7, 19\}$ under ring and hex-grid adjacency. It resists a training collapse that becomes total for a flattened-state single-agent baseline, converging on a residual 35.4% collapse rate across 64 independently-drawn seeds rather than either of two disagreeing small-sample estimates, a pattern already present at N=7 rather than only emerging at large cluster size. On the metric that reflects the actual training objective, the architecture decisively beats an independent-learner ablation that isolates the same coordination contribution, though its edge over the single-agent baseline specifically is small and not statistically decisive at this sample size, an honest limit we report. Federating the architecture costs nothing measurable in reward, and differential privacy degrades decision precision only past a real threshold rather than a smooth trade-off. Under disruption, its cost accelerates with severity like every arm tested, with a coordination-specific penalty at only the mildest severity tested.

These safeguards are not abstract, The correctness-aware pair diagnosed and fixed a real training collapse our first encoder suffered on all 30 campaign seeds (Section 4.1), and the reproduction-versus-replication discipline caught and retracted a real architectural overclaim under mid-episode disruption (Section 5.3). The live deployment below both confirms the offline-trained decision logic executes correctly on real hardware and also uncovered a cross-layer scheduler misconfiguration that had earlier manufactured the appearance of live contention. This was fixed at its root, after which we noted that the rig cannot exercise genuine per-slice scarcity at any load level tested, confining the disruption and scaling claims above to the disclosed offline stress regime where they are made.

2 Related Work

Our prior systematic review [9] screened 6,286 records down to 67 core 2024–2026 studies on machine-learning-driven, user-centric distributed network slicing, organised around three enabling pillars, namely topology-aware graph representation, distributed multi-agent control, and privacy-preserving federation, and found individual pillars well developed but rarely combined under realistic constraints, and never with a validated live deployment. None of the works we reviewed, there or here,

checks whether its own evaluation metric can mask a degenerate policy, or replicates a headline finding on an independent seed sample. This is the gap this paper fills.

The closest prior works run both checks against a cluster of three groups. Shao et al. [14] (the precedent behind our own GAT encoder choice, showing it strengthens inter-agent cooperation over plain DQN/Advantage Actor-Critic (A2C) baselines), Ahmadi et al. [2] (topology generalisation via graph attention, complementary to, not answered by, the fixed-topology evaluation-integrity question we ask in Section 5.4), and Sulaiman et al. [15] (the closest prior art to our own joint slicing-and-admission-control problem) together check neither whether their own reward-optimal policy could be mistaken for a degenerate one nor a headline result against an independent seed sample. Tashman and Cherkaoui [16] and Fatehi et al. [5] study robustness to a disruptive condition, as our own disruption campaign (Section 5.3) does, but neither checks whether the metric reporting that robustness is itself trustworthy or replicates a finding on an independent sample, exactly the gap our own campaign's independent-seed replication closes, and not hypothetically, since it is what overturned our initial churn-immunity reading. Zhang et al. [19] (the closest periodic-synchronisation precedent) and Yasin et al. [18] (the closest FL+DP precedent, targeting a different threat model, securing the controller rather than characterising a control policy's own privacy-utility trade-off, which our own σ-sweep in Section 5.2 reports) both lack a correctness-aware check on whether federating or privatising changed decision quality. A further, closely related set [3, 6, 12, 13] combines two of these three pillars but, like every work above, evaluates exclusively on its own reported outcome metric, computed once, on one seed or run. This is the gap this paper fills, not a new point in the GAT/MARL/FL design space but a demonstration (on GAT-CTDE, extended to federation, privacy, and disruption) that checking a metric's own trustworthiness and replicating a finding on an independent sample are both necessary, and that skipping either changes what a reader is entitled to believe the resulting numbers mean.

3 System Model and Problem Formulation

Fig. 2 orients the reader before the detail. A GAT-CTDE controller is exercised across four offline campaigns in a disclosed stress environment, every result passed through two correctness safeguards, and the offline-trained logic is then checked on the live single-gNB testbed of Fig. 3.

We consider a cluster of N=3 gNBs, each serving the same three slices URLLC, eMBB, and mMTC under a shared per-gNB Physical Resource Block (PRB) budget B. An admission decision is realised as a PRB-ceiling adjustment through the `slicing_control_m` min and max ratio rather than a literal per-flow accept or reject, since the real OAI xApp control API exposes only ratio ceilings. No real multi-gNB physical topology is available to us, so we take the least assumption-laden choice and treat the cluster as fully connected, noting this as a modelling choice rather than a measured fact. At each control step every gNB has zero or more pending admission requests, each tagged with a slice identity, and an agent decides accept or reject for each independently, conditioned on the current cluster state. The discrete action space

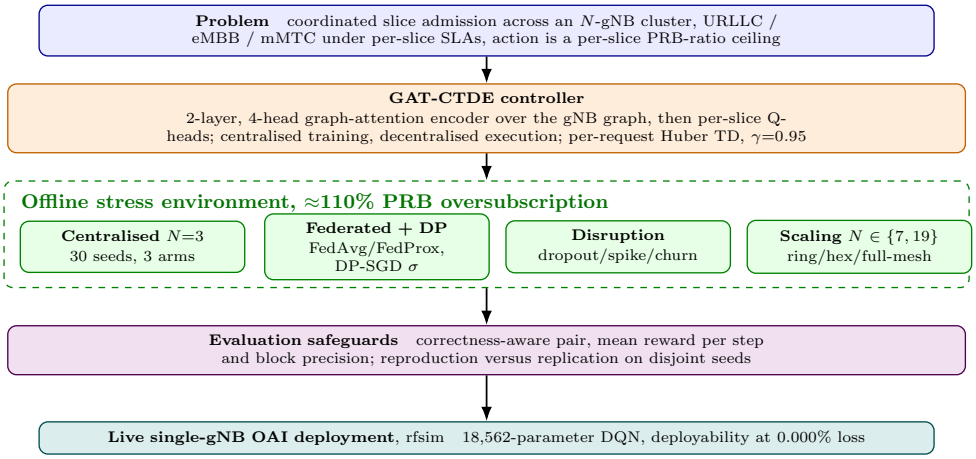


Fig. 2 Overview of the research. A GAT-CTDE controller is evaluated across four offline campaigns in a stress environment, every result checked by two safeguards, then validated on a live single-gNB testbed.

matches our single-gNB work exactly at `action_dim`=2, so any coordination benefit found here is attributable to the architecture and not a richer action space.

The step reward follows our prior single-gNB formulation, a per-slice service term net of SLA-violation penalties, a congestion charge on admitted volume, and a load-imbalance penalty across gNBs. The step reward r_t sums over slices indexed by k, with priority weight ω_k, per-accept reward R_k, admitted count n_k, violation penalty λ_k, and the indicator $\mathbf{1}[\cdot]$ equal to one exactly when slice k is in SLA violation that step,

$$r_t = \sum_k \big(\omega_k R_k\, n_k - \lambda_k\, \mathbf{1}[v_k]\big) \;-\; \kappa\, C_t \|a_t\|_1 \;-\; \beta\,(1 - F_t), \tag{1}$$

where $\|a_t\|_1$ is the total admitted this step, C_t the cluster-mean congestion level, κ the congestion coefficient, $F_t \in [0, 1]$ the min-over-max PRB-utilisation fairness ratio, and β the load-balance weight. The violation flag v_k derives from a continuous SLA margin m_k that the correctness-aware metrics of Section 4.4 also build on, taken as the worse of the queue and loss channels,

$$m_k = \min\Big(1 - \tilde{q}_k,\; 1 - \tfrac{\ell_k}{\ell_k^{\max}}\Big), \qquad v_k \Leftrightarrow m_k \le 0, \tag{2}$$

with $\tilde{q}_k$ the slice backlog relative to its budget and $\ell_k, \ell_k^{\max}$ its loss proxy and loss budget. These weights are tuned in prior work so that the reward-optimal policy under congestion is differentiated shedding, namely continue accepting URLLC, mostly accept eMBB, and reject mMTC, confirmed by Q-value inspection on a working baseline policy in Section 4.1 rather than assumed. Every result except Section 5.5's bounded live anchor is obtained in an offline, closed-loop simulation deliberately oversubscribed to roughly 110% of one gNB's PRB budget, reusing our existing multi-gNB configuration unmodified. We use this offline environment explicitly as a deliberately

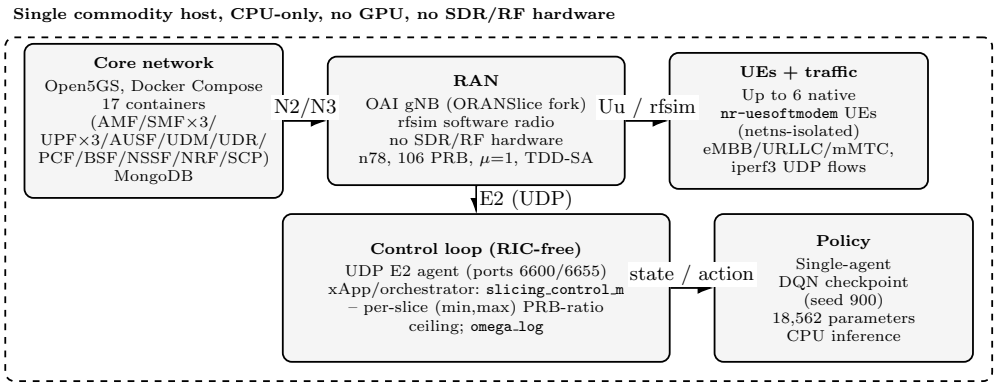


Fig. 3 Real testbed stack, a single-host, software-radio rfsim, RIC-free O-RAN deployment distinguished from SDR hardware-in-the-loop testbeds. Every block and label is a measured or configured value from this rig.

harder, disclosed stress environment for a contention regime the live rig cannot currently produce, since at every native load tested, one and two User Equipments (UEs) per slice alike, all three slices serve essentially 100% of offered demand with zero loss in Section 5.5. Multi-gNB and disruption claims therefore remain offline-only. The concrete rig underlying every live measurement reported later is the single-host, software-radio, RIC-free O-RAN stack of Fig. 3, which the live deployment of Section 5.5 exercises end to end.

4 GAT-CTDE Method

4.1 Shared Encoder, Decentralised Q-Heads

Fig. 4a shows the architecture. At every control step, each gNB's own per-slice features, namely PRB-used-ratio, congestion-level, and queue-length-norm, form that node's row x_i in a joint node-feature matrix $X \in \mathbb{R}^{N\times d}$ over the N gNB nodes with d features each. A two-layer, H=4-head GAT [17] produces one embedding per gNB. Rather than scoring a concatenation, each head h decomposes additive attention into a learned source and destination vector a_s^h, a_d^h applied to the shared linear projection W^h, with ϕ the leaky rectifier of slope 0.2 and $\mathcal{N}_i$ the neighbours of node i under the adjacency mask,

$$\varepsilon_{ij}^h = \phi\big(a_s^{h\top} W^h x_i + a_d^{h\top} W^h x_j\big), \qquad \alpha_{ij}^h = \frac{\exp \varepsilon_{ij}^h}{\sum_{l\in\mathcal{N}_i} \exp \varepsilon_{il}^h}. \tag{3}$$

Here ε_{ij}^h is the unnormalised attention score from node i to neighbour j in head h, and α_{ij}^h its normalisation over i's neighbours, with l ranging over $\mathcal{N}_i$. Hidden layers concatenate the H heads under an ELU nonlinearity, while the final layer averages them, and a per-layer normalisation operator LN$(\cdot)$ that rescales each node's embedding to

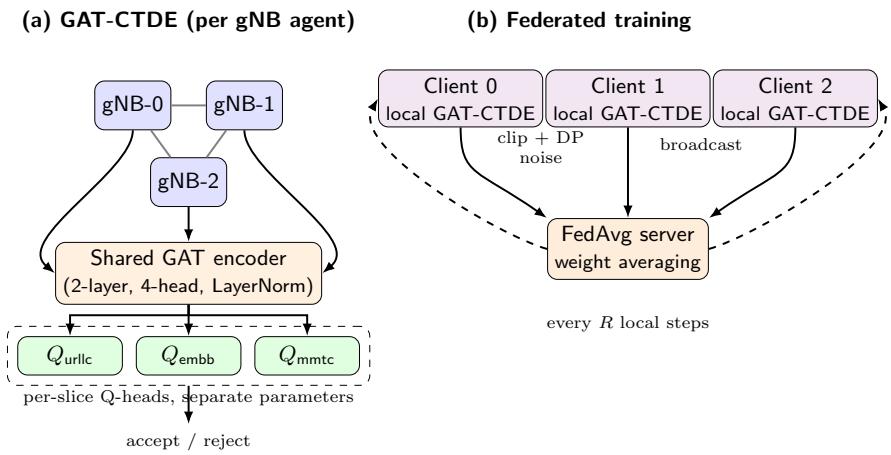


Fig. 4 GAT-CTDE architecture. a, A shared graph-attention encoder over the fully-connected 3-gNB topology produces one embedding per node, consumed at execution time by per-slice Q-heads using only that node's row. b, The federated variant, each gNB a client synchronised by periodic FedAvg with an optional DP-SGD clip-and-noise step.

zero mean and unit variance produces the final per-gNB embedding z_i,

$$z_i = \mathrm{LN}\Big(\tfrac{1}{H}\textstyle\sum_{h=1}^{H}\sum_{j\in\mathcal{N}_i}\alpha_{ij}^{h}\,W^{h}x_j\Big). \tag{4}$$

Centralisation comes from this encoder, whose parameters are updated by every agent's loss jointly so that it sees the full cluster state during training. Decentralised execution comes next, since at action-selection time each agent consumes only its own embedding row z_i to decide its own pending requests.

Both the `LayerNorm` and the per-slice heads below were made necessary by a real training collapse rather than added on theoretical grounds. An early design fed each embedding, concatenated with a one-hot slice context, into a single shared Q-head multi-layer perceptron. With all three slices updating the same shared parameters, URLLC's large priority weight produces proportionally large gradients that overwrite mMTC's much smaller true signal, a standard shared-parameter class imbalance that drove the evaluation policy to block zero requests. Replacing the single shared head with one small perceptron per slice, selected by the context one-hot, gives each slice its own parameters so one slice's gradients can no longer overwrite another's. Together with the `LayerNorm`, this reduced the collapse from 30 of 30 to 9 of 30 seeds, with correctly-targeted differentiated shedding rising from 0 to 21 of the 30 seeds. A seed here is one independent random initialisation of network weights and environment stream, and we report every seed rather than a single best run.

Algorithm 1 states the training loop. The encoder is updated from every request in a sampled minibatch regardless of gNB or slice, which is the centralised training, while action selection at rollout time consults only the acting agent's own embedding row, which is the decentralised execution. Every pending request at gNB i, slice k is its own temporal-difference sample, never summed across a step before the loss. For a

Algorithm 1 GAT-CTDE: centralised training, decentralised execution

Require: encoder f_θ, per-slice Q-heads $\{Q_{\phi_k}\}$, targets $f_{\theta^-}, \{Q_{\phi_k^-}\}$, buffer $\mathcal{D}$, adjacency A, discount γ, sync period C

for each step t, observing joint node features X_t **do**
  $E_t \leftarrow f_\theta(X_t, A)$ {centralised, one row per gNB}
  for each pending request at gNB i, slice k, pick $a \leftarrow \arg\max_a Q_{\phi_k}(E_t[i])[a]$, ϵ-greedy {decentralised, own row only}
  step env, store $(X_t, X_{t+1}, \{(i,k,a)\}, r_t, \text{done})$ in $\mathcal{D}$
  if $|\mathcal{D}| \geq$ warmup **then**
    sample minibatch B; $E_B \leftarrow f_\theta(X_B, A)$, $E'_B \leftarrow f_{\theta^-}(X'_B, A)$
    per request, $y \leftarrow r + \gamma(1-\text{done})\max_{a'} Q_{\phi_k^-}(E'_B[i])[a']$
    $\mathcal{L} \leftarrow$ mean per-request Huber loss, never summed per step; update $\theta, \{\phi_k\}$, clip grad-norm 1.0
  end if
  every C episodes, $\theta^- \leftarrow \theta$, $\phi_k^- \leftarrow \phi_k$
end for

request with immediate reward r, next-state embedding z'_i, and candidate next action a', the bootstrapped target y at discount γ and the minibatch loss are

$$y = r + \gamma\,(1-d)\max_{a'} Q_{\phi_k^-}(z'_i)[a'], \qquad \mathcal{L} = \frac{1}{|B|}\sum_{(i,k,a)\in B} \ell_\delta\Big(Q_{\phi_k}(z_i)[a] - y\Big), \tag{5}$$

where Q_{ϕ_k} is slice k's online Q-head and $Q_{\phi_k^-}$ its target copy, d the episode-done flag, B the sampled minibatch of requests, and ℓ_δ the Huber loss, used in place of squared error to keep the deeper encoder stable under long bootstraps.

4.2 Federated Training with Differential Privacy

Fig. 4b shows the federated variant, in which each gNB becomes a federated-learning client holding a private copy of the same network and training only on its own requests. Transitions are partitioned by owning agent, so no client sees another's raw admission data, while its GAT encoder still consumes the full joint node-feature snapshot. Every R local steps the server averages all clients' online weights by FedAvg [11] and broadcasts the result back. We also implement FedProx [10], which adds a proximal term pulling local weights toward the last broadcast global model. Swept at $\mu \in \{0.01, 0.1, 1.0\}$ under both homogeneous and heterogeneous per-gNB load, FedProx earns no measurable dividend at any strength, since per-round client drift is negligible relative to the loss scale.

Differential privacy, a formal guarantee that one client's training data cannot be inferred from the shared model, is applied as a standard DP-SGD [1] step. Every client's gradient is clipped to a fixed norm C_{dp} and, when enabled, perturbed with Gaussian noise scaled by a noise multiplier σ before that client's optimiser step. We report σ as the primary privacy parameter alongside a formal (ε, δ) bound via zero-concentrated differential privacy [4], where ε and δ are the standard privacy-loss and

Algorithm 2 Federated GAT-CTDE round with DP-SGD

Require: N clients with local nets $(\theta_i, \{\phi_{i,k}\})$, global snapshot $\bar{\theta}$, local steps R, clip norm C_{dp}, noise σ, FedProx weight μ

for each round **do**
  for R local steps, each client i independently **do**
    $\mathcal{L}_i \leftarrow$ per-request Huber TD loss of Alg. 1 on i's own requests $+ \frac{\mu}{2}\|\theta_i - \bar{\theta}\|^2$
    $g_i \leftarrow \text{clip}(\nabla\mathcal{L}_i, C_{\text{dp}})$, then $g_i \leftarrow g_i + \mathcal{N}(0, (\sigma C_{\text{dp}})^2 I)$ if $\sigma > 0$ {same clip isolates privacy cost}
    $(\theta_i, \phi_i) \leftarrow (\theta_i, \phi_i) - \eta\, g_i$
  end for
  $\bar{\theta}, \bar{\phi} \leftarrow \frac{1}{N}\sum_i (\theta_i, \phi_i)$ and broadcast to every client {FedAvg}
end for

failure-probability parameters. Composing the Gaussian mechanism over m per-client gradient releases gives the zCDP parameter ρ and the corresponding bound

$$\rho = \frac{m}{2\sigma^2}, \qquad \varepsilon(\delta) = \rho + 2\sqrt{\rho \ln(1/\delta)}. \tag{6}$$

Clipping is applied identically whether or not noise is added, so a σ=0 run and a σ>0 run differ only in whether noise is injected, the isolation the privacy-cost comparison in Section 5.2 depends on. Algorithm 2 states one federated round, with proximal weight μ the FedProx strength swept above.

4.3 Experimental Setup

Both campaigns use 300 training and 50 evaluation episodes per seed. The centralised campaign uses a fixed 30-seed list, seeds 900–929, across three arms, namely GAT-CTDE, a single-agent DQN baseline run against the flattened joint state, and an independent-DQN ablation. An ablation is a deliberately reduced variant with one component removed to isolate its effect, here one per-gNB DQN with no parameter sharing and no topology information, which isolates the contribution of the shared encoder and centralised training. All three arms use matched hyperparameters, with discount factor $\gamma = 0.95$ and per-episode epsilon decay. The federated campaign uses the first 10 of these seeds and a five-level noise sweep $\sigma \in \{0, 0.5, 1, 2, 4\}$, so its σ=0 level is directly comparable to the same 10 centralised seeds. All confidence intervals are 95% bootstrap percentile intervals over 10,000 resamples, and paired comparisons use the Wilcoxon signed-rank test.

4.4 Evaluation Safeguards

Two safeguards make the results below trustworthy, and Table 1 summarises the four pipeline failures they caught, each of which a standard pipeline would have surfaced as a clean result. The first safeguard is a correctness-aware metric pair computed from data every run already logs. Mean reward per step is the actual training objective, and block precision is the fraction of blocked requests correctly targeting mMTC, the

Table 1 Four evaluation-pipeline failures this paper's safeguards caught, none of which a standard pipeline would have surfaced.

Failure	Missed by	Caught by	Outcome
Training collapse (§4.1)	Compliance blind to no-op	Correctness-aware pair	0/30→21/30 differentiate
Eval-log contamination	Append-mode log stacking	Episode-index audit	Logs cleared, no retrain
Reproducibility gap (§5.3)	Single-sample overclaim	Repro-vs-replication	Churn claim retracted
Scheduler-floor bug (§5.5)	Can't see cross-layer floor	Direct state instrumentation	>100k readings clear

only slice the reward calibration ever makes reject-optimal. Writing b_{mMTC} and b for the mMTC-targeted and total blocks over an evaluation,

$$\pi = \frac{b_{\mathrm{mMTC}}}{b}, \quad b > 0, \qquad \text{undefined for } b = 0, \tag{7}$$

so a seed that never blocks is excluded rather than scored as a success.

These are needed because `sla_compliance_all_slices` scores worse when the policy becomes more correct, since blocking mMTC under congestion never recovers mMTC's own SLA margin in this stress regime. Re-checking every paired comparison in this paper against both metrics, the correctness-aware pair and compliance disagree on direction or significance in 44 of 49 comparisons, 89.8%, confirmed at 21 of 24 on a disjoint supplementary seed range. Every number in this paper that matters is therefore reported through the correctness-aware pair.

The second safeguard is a reproduction-versus-replication discipline, where every headline comparison is recomputed on an independent, disjoint seed sample drawn after the first and both are reported together. Reproduction re-runs the same seeds, whereas replication draws fresh ones, so only replication can catch a result that is an artifact of one seed sample. Applied to the disruption campaign in Section 5.3, it corrected an early reading that independent DQN was immune to agent churn, which the committed sample alone had made look confirmed. The remaining two failures were a logging bug that had stacked three runs' evaluation episodes in each seed's log, cleared and re-evaluated against the already-trained checkpoints with no retraining, and a live scheduler-floor misconfiguration resolved in Section 5.5.

5 Results

5.1 Centralised

The N=3 fully-connected topology is deliberately the setting where a shared graph-attention encoder has the least topology to exploit, which makes it a clean substrate for the evaluation-integrity findings, uncomplicated by any topology effect. The encoder's coordination advantage is tested properly only once the cluster scales to N=19 under sparser adjacency in Section 5.4.

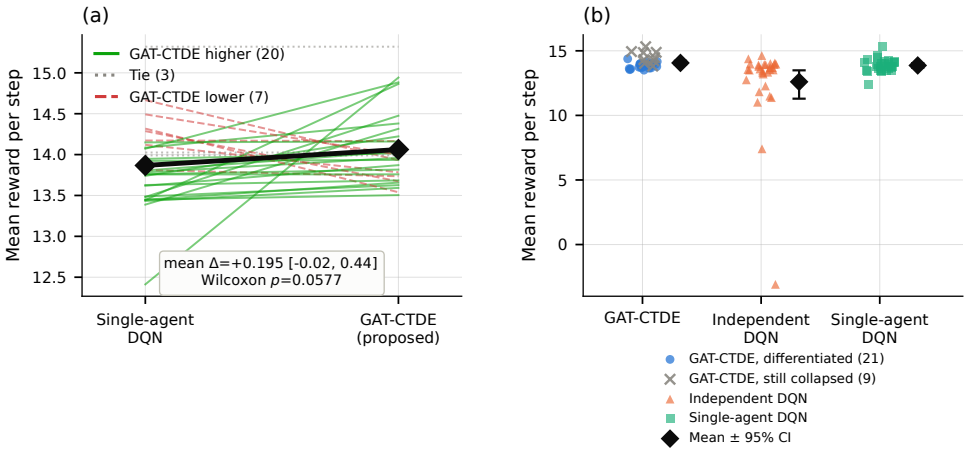


Fig. 5 M2 centralised campaign, per-seed evidence. a, Paired per-seed mean reward per step versus single-agent DQN, coloured by win, tie, or loss, the source of the borderline $p = 0.0577$. b, Per-seed reward distribution for all three arms.

Table 2 M2 campaign results, 30 seeds/arm, 95% bootstrap CIs.

Arm	Mean reward/step	Block precision	Compliance*
GAT-CTDE	14.062 [13.91, 14.23]	0.952	0.151 [0.083, 0.225]
Indep. DQN	12.601 [11.30, 13.49]	0.901	0.049 [0.013, 0.093]
Single-agent	13.867 [13.69, 14.05]	0.987	0.115 [0.061, 0.176]

* Low compliance is the metric artifact Section 4.4 explains, not a performance failure.

Table 2 and Fig. 5 report the final 30-seed campaign. On mean reward per step, the metric that reflects the actual objective, GAT-CTDE beats the independent-DQN ablation decisively by +1.461, 95% CI [0.550, 2.782] and $p < 0.0001$, isolating a real contribution from the shared encoder and centralised training. Its edge over single-agent DQN is small and not statistically decisive at this sample size, +0.195 with 95% CI [−0.017, 0.442] and $p = 0.0577$. Block precision is high at 0.952, so when GAT-CTDE blocks a request it is almost always the reward-correct slice. The compliance metric shows no clear direction at +0.036 with $p = 0.4549$, the artifact Section 4.4 explains.

5.2 Federated Training and DP

Comparing the centralised campaign against the federated variant with no DP noise on the same 10 seeds, mean reward per step moves by +0.133, 95% CI [−0.138, 0.470] and $p = 0.8125$, no measurable cost to federating, and both arms hold perfect block precision on seeds that block at all. Sweeping the DP noise multiplier in Fig. 6b, block precision is perfect through σ=1.0 then drops sharply to 0.834 at σ=2.0 and 0.584

Table 3 Training hyperparameters, all fixed across seeds.

Hyperparameter	Value
Learning rate (GAT-CTDE)	1×10^{-4}
Learning rate (DQN baselines)	1×10^{-3}
Discount γ	0.95
Epsilon: start $\rightarrow$ end (decay/episode)	1.0 $\rightarrow$ 0.05 (0.985)
Target-network sync	every 10 episodes
Optimiser	Adam
Training / evaluation episodes	300 / 50
GAT encoder, GAT-CTDE only	
Hidden / output dim	16 / 16
Attention heads	4
Encoder layers	2
Q-head hidden dim	32
Federated + DP, federated GAT-CTDE only	
Local steps per round	50
Aggregator	FedAvg (μ=0)
DP gradient clip norm	1.0
DP noise multiplier sweep σ	$\{0, 0.5, 1, 2, 4\}$

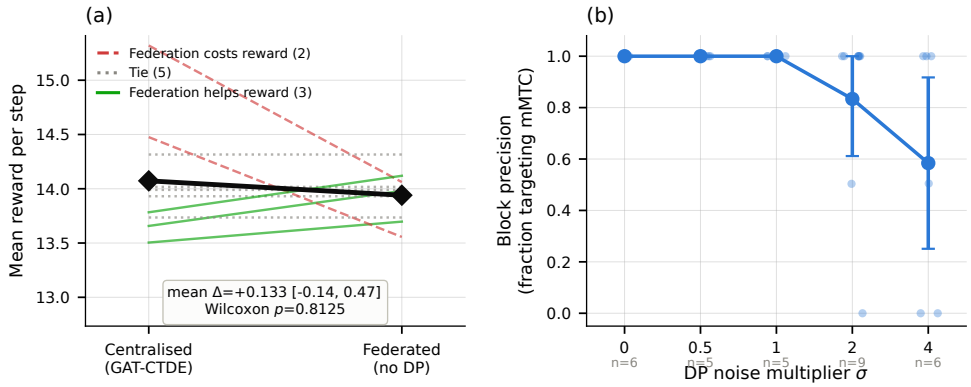


Fig. 6 M3 federated and DP sweep. a, Paired per-seed federation cost on mean reward per step, no measurable difference at $p = 0.8125$. b, Block precision versus DP noise multiplier σ, perfect through σ=1.0 then dropping sharply.

at σ=4.0, a real threshold rather than a smooth degradation, while mean reward per step stays within a narrow 12.45–14.19 band, since DP noise erodes which slice is blocked rather than whether the policy differentiates at all. Read with the FedProx null result, this answers a deployability question directly, namely that periodic, infrequent synchronisation suffices for this architecture, so a disaggregated RAN Intelligent Controller needs no per-step cross-gNB coordination to match centralised reward, and differential privacy can be added at a bounded, known precision cost.

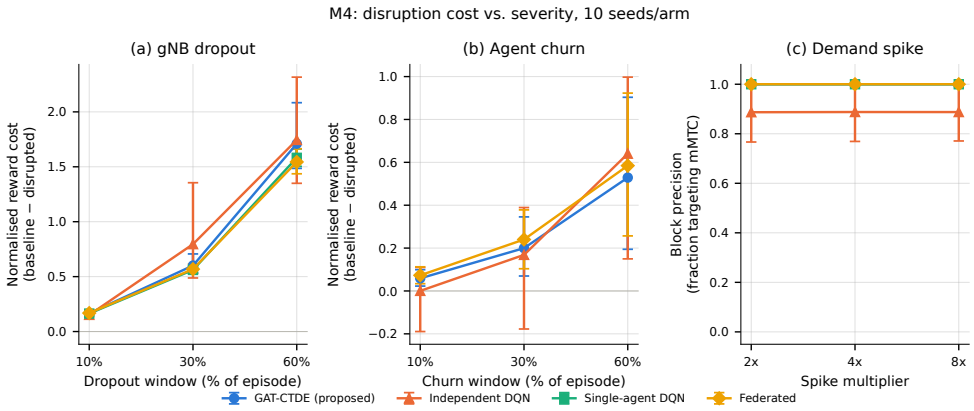


Fig. 7 M4 disruption campaign, 10 seeds per arm. a, b, Normalised reward cost versus severity for dropout and churn, accelerating with severity. c, Block precision versus demand-spike multiplier, flat and near-ceiling.

5.3 Disruption

We evaluate every arm's frozen checkpoint, with no retraining, against three mid-episode disruptions injected at evaluation time. 'gNB dropout' forces a gNB's requests to reject and zeroes its row in the joint feature matrix for a window. 'Demand spike' is a transient arrival-rate multiplier external to the policy. 'Agent churn' substitutes a gNB's decisions with a fresh, never-trained copy of its own policy class for a window, leaving other agents untouched, and applies to every genuinely multi-agent arm but not single-agent DQN. Severity is the window duration, 10%, 30%, or 60% of the episode, for dropout and churn, and the arrival-rate multiplier, 2×, 4×, or 8×, for spike. Since raw reward inflates mechanically with request volume, we report the volume-normalised mean reward $\bar{r}$ over an episode of T steps, dividing each step reward r_t by its own pending-request count n_t before averaging,

$$\bar{r} \;=\; \frac{1}{T}\sum_{t=1}^{T}\frac{r_t}{\max(n_t,\,1)}. \tag{8}$$

Every dropout and churn cost accelerates with severity rather than growing proportionally, with Wilcoxon $p \leq 0.03$ and 9 or 10 of 10 seeds hurt in every case. On the committed sample GAT-CTDE's dropout cost rises 0.165, 0.601, 1.710 across the three severities and its churn cost 0.059, 0.200, 0.529, with every other arm showing the same accelerating shape. Demand spike does not corrupt decision quality on the volume-invariant block-precision metric in Fig. 7c, so its apparent per-step reward gain is a denominator artifact rather than better decisions.

Independent DQN's churn cost needs both seed samples to state honestly. The committed sample is small and not conventionally significant, which an earlier pass read as independent DQN being immune to churn, since it never attends to another

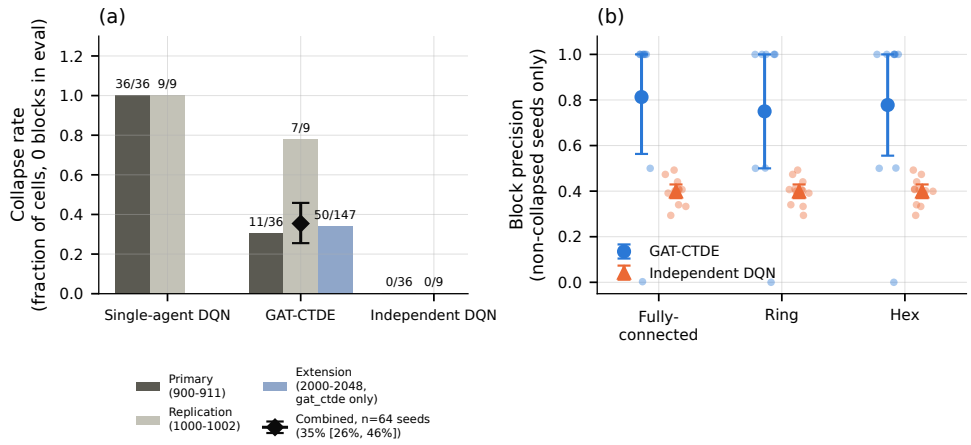


Fig. 8 N=19 cluster, per-seed evidence. a, Collapse rate pooled across topologies, single-agent DQN total and GAT-CTDE converging to 35.4% across 64 seeds. b, Block precision by topology for non-collapsed seeds.

agent's features. A disjoint independent-seed replication is significant at every severity and larger than the coordinated arms', its cost rising 0.097, 0.657, 1.223 against the committed 0.000, 0.168, 0.640, so the corrected reading is that churn always hurts and treating a borderline n=10 result as a confirmed property was the error. This is the one exception among four cross-section headline comparisons checked against an independent sample. The other three replicate with overlapping intervals, namely GAT-CTDE over independent DQN on reward at +1.461 then +0.767 with $p < 0.0001$ in both, GAT-CTDE over single-agent at +0.195 then +0.445, and a federation cost negligible in both.

5.4 Cluster-Size Scaling

We extend the cluster from N=3 to $N \in \{7, 19\}$ under three adjacency structures, fully-connected, ring, and hex-grid, scaling synthetic arrivals with N to hold per-gNB offered load roughly constant. GAT-CTDE is the only arm whose Q-head consumes this adjacency, so independent DQN and single-agent DQN serve as a topology-invariance check. We normalise reward per gNB as in Eq. 8 and report block precision alongside, since compliance is OR-aggregated across gNBs and trends to zero as N grows regardless of policy quality. The primary sample is 12 seeds at N=19 across all topologies and arms, with independent replication seeds for generalisation.

At N=19 the paired per-gNB reward margin is not significant for any topology, which is mechanically expected since single-agent DQN's always-accept collapse collects the service term on every request while GAT-CTDE pays a real cost each time it correctly rejects mMTC. The advantage therefore lives in collapse resistance and block precision.

Single-agent DQN collapses totally at N=19, 15 of 15 cells combined across both samples and every topology, confirmed at the individual-decision level by a direct Q-value probe. GAT-CTDE collapses only partially, and independent DQN never fully collapses but sits in a narrow, mediocre 0.29–0.49 block-precision band identical across topologies, as expected for an arm that never reads adjacency.

GAT-CTDE's collapse rate converges, across 64 independently-drawn seeds pooling 68 of 192 cells, to

$$\hat{p}_{\text{collapse}} = 35.4\%, \qquad 95\%\ \text{CI}\ [25.5\%,\ 45.8\%], \tag{9}$$

a single seed-level-bootstrapped estimate rather than either of two smaller disagreeing samples. A properly powered 12-seed resample at N=7 under the identical stress environment, with a disjoint 12-seed replication, puts GAT-CTDE at 41.7% and 58.3% collapse across the two samples, statistically indistinguishable from Eq. 9, so its collapse-proneness is present from N=7 onward rather than emerging only at large cluster size. Over the same two samples single-agent DQN collapses at 91.7% and 75.0% against its earlier three-seed pilot's 0%, the clearest single result in this extension, and independent DQN at 0.0% and 16.7%, confirming it collapses rarely rather than never.

5.5 Live Deployment

This section reports what the real OAI testbed of Fig. 3, introduced in Section 3, can and cannot establish about the offline-trained policy, using a single-agent-DQN checkpoint, seed 900, against a real gNB, real UEs, and real per-slice User Datagram Protocol (UDP) traffic at native load of 1 UE per slice and double native load of 2 UEs per slice.

A live capacity probe at a deliberately wide-open ceiling found all three slices serve essentially 100% of offered demand with 0.000% loss at both loads. The frozen checkpoint's decision logic executes correctly end to end, with real traffic through the real E2 loop, real accept and reject decisions, and its characteristic low-priority-only blocking of mMTC reproducing at both loads. This establishes deployability. Since the rig never reaches scarcity, the pattern is equally consistent with the policy's trained priority bias, so admission-control-under-contention is claimed only in the disclosed offline stress regime, not live. An early scheduler-configuration issue, where `saclb_live.yaml`'s per-slice PRB-ratio ranges converted to raw grants below the gNB scheduler's hard 5-PRB minimum, was fixed at its root, after which direct instrumentation confirmed zero below-floor grants across more than 100,000 readings. A separate unit-mismatch in the live SLA-margin diagnostic is fixed by converting backlog to a queueing-delay estimate via Little's Law against each slice's own latency budget.

6 Conclusion

We set out to test whether a standard evaluation metric and a live testbed's evidentiary reach can be trusted in multi-agent, privacy-constrained, disruption-prone slice admission control. A correctness-aware metric pair, a reproduction-versus-replication

discipline, and direct live instrumentation together surfaced four pipeline failures a standard pipeline would have reported as clean results, and every headline number here is reported through those safeguards.

On the metric that reflects the training objective, GAT-CTDE beats an independent-learner ablation decisively, though its edge over the single-agent baseline is small and not decisive at this sample size. Federation adds no measurable reward cost, differential privacy is costly only past a real precision threshold, and disruption cost accelerates with severity for every arm. Scaling to 19 gNBs, single-agent DQN collapses totally while GAT-CTDE remains only partially collapse-prone, converging across 64 seeds on a 35.4% rate with 95% CI [25.5%, 45.8%] already present at 7 gNBs. A live single-gNB deployment confirms the offline-trained logic executes correctly on real hardware, establishing deployability, with admission-control-under-contention claimed only in the disclosed offline stress regime since this rig cannot yet reach genuine per-slice scarcity.

The main limits follow from that scope. Every multi-gNB, federated, and disruption result is offline, the stress environment is deliberately harder than any load this rig reaches, the graph models only the gNB cluster with no edge-compute dimension, and the federated setting is a single-operator privacy scenario. The broader lesson is that evaluation in these settings needs metrics tied to the training objective, architectural claims need an independent seed sample before they are trusted, and a live testbed's reach must be checked rather than assumed.

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Declarations

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- **Competing interests:** The authors declare no competing interests.
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- **Consent for publication:** Not applicable, since this manuscript contains no personal data or identifying information about any individual.
- **Data availability:** The experiment logs, checkpoints, and campaign result files underlying this paper's findings are available from the corresponding author on reasonable request.
- **Materials availability:** Not applicable, since this study involves no physical or biological materials.

- **Code availability:** The framework and experiment scripts used to produce this paper's results, including the reproducibility manifest and analysis scripts used to independently re-derive every table and figure in this paper, are available from the corresponding author on reasonable request.
- **Author contribution:** M. P. Kunasegran: conceptualisation, methodology, software, investigation, writing (original draft). W. L. Pang: supervision, writing (review and editing). S. K. Phang: supervision, writing (review and editing).

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